



Designing membrane-targeted peptides requires understanding sequence--environment interactions. This thesis combines molecular simulation, biophysical modelling, and computational optimization to investigate these interactions. Free-energy calculations reveal how membrane composition and peptide orientation influence binding, while complementary models incorporate environmental effects into peptide conformational and sequence design within an α -helical scaffold. Classical and hybrid quantum--classical algorithms explore these formulations, linking peptide--membrane interactions with environment-conditioned sequence design. Together, these contributions provide a framework to identify peptides with specific environmental preferences and prioritize candidates for simulation and experimental validation.